Problem 1 – Software Development Project

The following activities are involved in a software development project:

Activity	Predecessor	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	—	2	4	6
B	A	3	5	7
C	A	2	3	4
D	B	4	6	8
E	C	3	5	7
F	D, E	2	4	6

Questions:

1. Calculate the expected time and variance for each activity.
2. Draw the PERT network.
3. Identify all possible paths.
4. Determine the critical path.
5. Calculate the expected project completion time.
6. Calculate the variance and standard deviation of the project duration.

Problem 2 – Web Application Project

A web application development project consists of the following activities:

Activity	Predecessor	a	m	b
A	—	1	3	5
B	A	3	5	9
C	A	2	4	6
D	B	4	6	8
E	C	3	4	7
F	D, E	2	3	4
G	F	1	2	3

Questions:

1. Calculate t_e and variance for each activity.
2. Construct the project network.
3. Find the expected duration of each path.
4. Determine the critical path.
5. Calculate the expected project duration.

6. Calculate the project standard deviation.
7. Determine the probability that the project will be completed within **30 days**.

Problem 3 – Online Banking System

An online banking system is developed through the following activities:

Activity	**Predecessor**	**a**	**m**	**b**
A – Requirements	—	2	4	6
B – Analysis	A	3	5	7
C – System Design	A	4	5	8
D – Database Design	B	2	4	6
E – Interface Design	C	3	5	7
F – Coding	B, C	5	7	9
G – Integration	D, E, F	2	4	6
H – Testing	G	3	5	7

Questions:

1. Calculate the expected time and variance of every activity.
2. Draw the PERT network.
3. Calculate the expected duration of all possible paths.
4. Identify the critical path.
5. Calculate the expected project duration.
6. Calculate the standard deviation of the critical path.
7. Determine the probability of completing the project within **35 days**.

Problem 4 – Mobile Application Development

A mobile application project has the following activities:

Activity	**Predecessor**	**a**	**m**	**b**
A	—	2	3	4
B	A	4	6	8
C	A	3	5	7
D	B	3	5	9
E	C	2	4	6
F	B, C	4	6	8
G	D, F	2	4	6

Activity	Predecessor	a	m	b
H	E, G	3	4	5

Questions:

1. Calculate the expected time and variance for each activity.
2. Determine all possible paths.
3. Calculate the expected duration of each path.
4. Identify the critical path.
5. Calculate the expected project completion time.
6. Calculate project variance and standard deviation.
7. Find the probability of completing the project within **32 days**.

Problem 5 – Hospital Management System

A Hospital Management System is being developed with the following activities:

Activity	Predecessor	a	m	b
A – Requirement Collection	—	2	4	6
B – Requirement Analysis	A	3	5	7
C – Database Design	A	2	3	4
D – System Design	B	4	6	8
E – Database Development	C	3	5	9
F – Module Development	B, D	5	7	11
G – Integration	D, E, F	2	4	6
H – Testing	G	3	5	7
I – Deployment	H	1	2	3

Questions:

1. Calculate the expected time and variance for each activity.
2. Construct the PERT network.
3. Determine the expected duration of each path.
4. Identify the critical path.
5. Calculate the expected project completion time.
6. Calculate the project variance and standard deviation.
7. If the target completion time is **40 days**, calculate the probability of completing the project within the target time.